DIGITAL DOOR ACCESS SYSTEM

USING COMBINATIONAL LOGIC

By:

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INTRODUCTION

- Mechanical keys can be lost, copied or picked, and many electronic locks are complex and costly for simple needs.
- This project uses basic gates (AND, OR, NOT, XOR) to unlock a door only when the entered digital code matches the authorised code.
- Purely combinational: the output depends only on the present inputs, with no memory or clock.
- Designed with Boolean algebra, truth table and K-map, then simulated in Logisim.
- Use cases: homes, laboratories, offices and other restricted areas.

OBJECTIVES

- Design a combinational circuit that unlocks only for authorised codes.
- Derive and minimise the Boolean expression.
- Verify the design with a truth table and K-map.
- Simulate correct, incorrect and edge cases in Logisim.
- Add zone-based control to save energy.

COURSE OUTCOMES (COS)

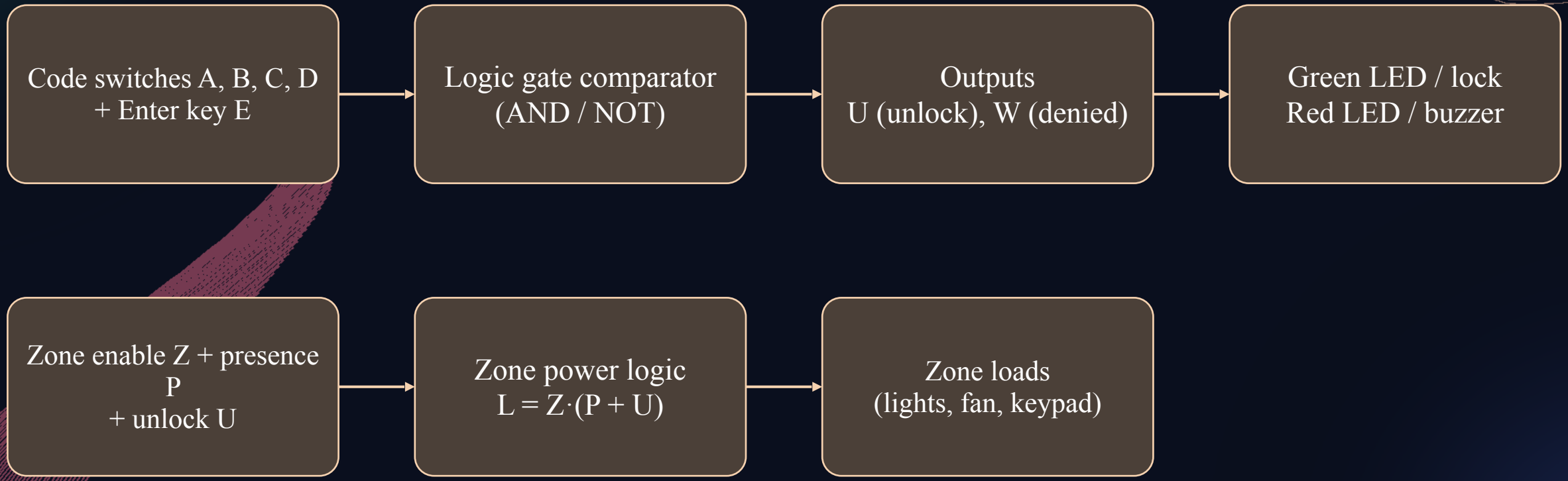
- CO1: Apply Boolean algebra and logic gates to digital circuits.
- CO2: Design combinational circuits for a given specification.
- CO3: Use truth tables and K-maps to minimise Boolean expressions.
- CO4: Simulate, test, and troubleshoot digital circuits.
- CO5: Explain the real-time application of digital electronics.

COMPONENTS REQUIRED

Component	Function
Input switches (A, B, C, D)	Enter the 4-bit access code (A is the MSB)
Enter key (E)	Triggers the code check; E = 0 keeps door locked, no alarm
Presence sensor (P)	Detects whether someone is in a zone
Zone enable (Z)	Allows a zone to be powered
AND / NOT / OR gates	Compare code with 1011 and 1111; produce U, W and zone power L
Green LED / door lock	Shows access granted (U = 1)
Red LED / buzzer	Shows access denied (W = 1)
Zone loads, power supply	Lights, fan, keypad; 5V DC supply

Software: Logisim for circuit design and simulation.

BLOCK DIAGRAM



The comparator checks the code against 1011 (user) and 1111 (master). With E = 0 the door stays locked and no alarm sounds.

CIRCUIT WORKING

- The user sets the 4-bit code on switches A, B, C, D and presses Enter (E = 1).
- AND and NOT gates compare the code with the authorised codes 1011 and 1111.
- Match: U = 1, so the door unlocks and the green LED turns ON.
- No match: W = 1, so the door stays locked and the red LED / buzzer turns ON.
- Zone logic L = Z·(P + U) powers a zone only when enabled and occupied or just unlocked.

TRUTH TABLE (E = 1)

A	B	C	D	U	W
0	0	0	0	0	1
0	0	0	1	0	1
0	0	1	0	0	1
0	0	1	1	0	1
0	1	0	0	0	1
0	1	0	1	0	1
0	1	1	0	0	1
0	1	1	1	0	1

A	B	C	D	U	W
1	0	0	0	0	1
1	0	0	1	0	1
1	0	1	0	0	1
1	**0**	**1**	**1**	**1**	**0**
1	1	0	0	0	1
1	1	0	1	0	1
1	1	1	0	0	1
1	**1**	**1**	**1**	**1**	**0**

Only 1011 (user) and 1111 (master) give U = 1. With E = 0, every code gives U = 0 and W = 0 (locked, no alarm).

LOGIC DESIGN

Authorised codes: 1011 = A·B'·C·D, 1111 = A·B·C·D

K-map: B is a don't-care, so

U = E · A · C · D

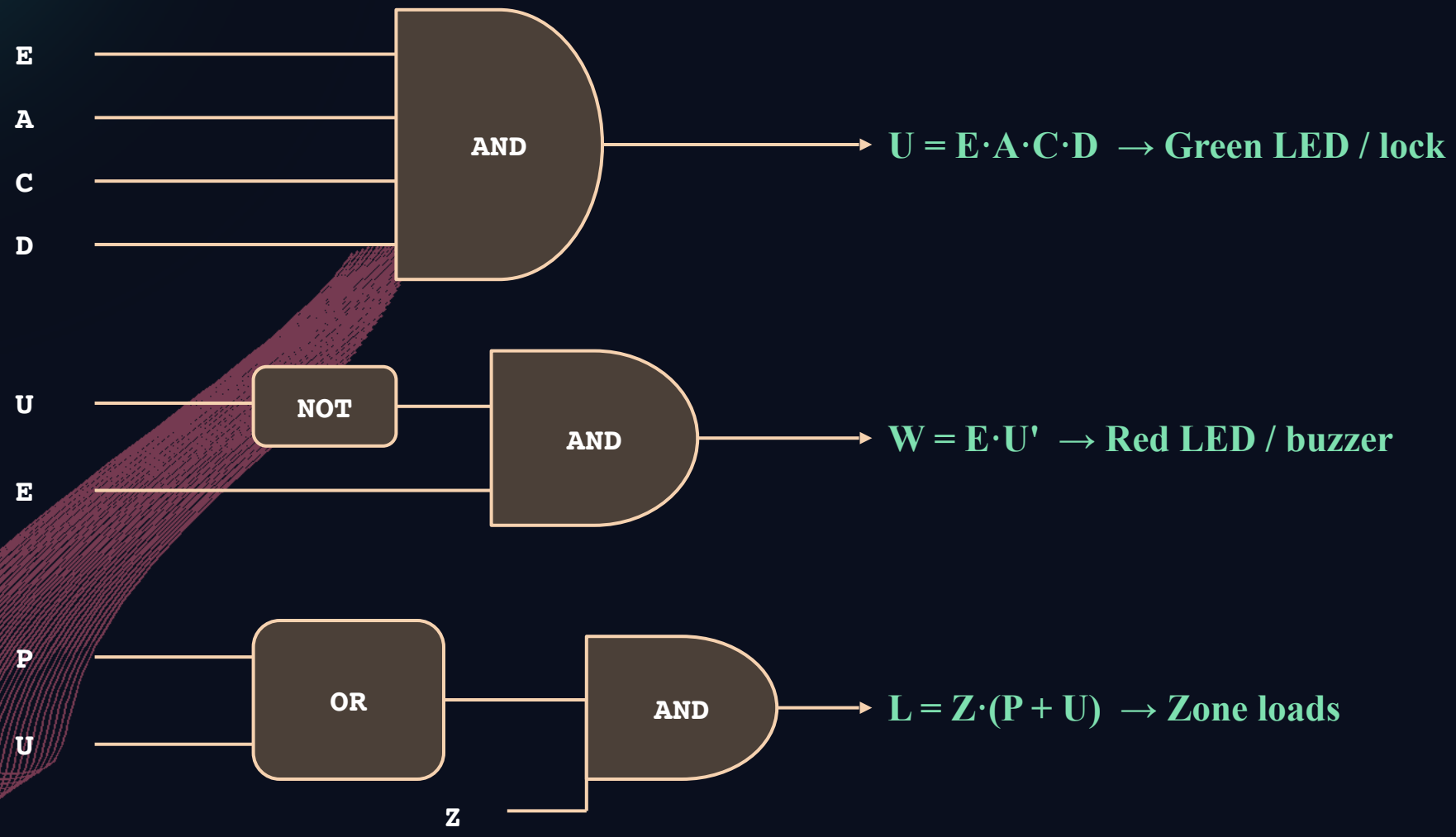
Denied: W = E · U'

Zone power: L = Z · (P + U)

AB \ CD	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	0	0	**1**	0
10	0	0	**1**	0

K-map of U (E = 1): group the two 1s (1011, 1111).

LOGIC DIAGRAM



SIMULATION / IMPLEMENTATION

- Design and simulate the circuit in Logisim (Tinkercad or Proteus also work).
- Use input pins for A, B, C, D, E, P, Z and LEDs for U, W and L.
- Test all 16 codes with E = 1, then with E = 0.
- Verify only 1011 and 1111 give U = 1; near-miss codes such as 1001 and 1110 are denied.
- Check zone logic: idle zones stay OFF.

ADVANTAGES

- Simple and easy to understand.
- Quick response: no clock or memory delay.
- Low cost and simple hardware.
- Easy to simulate and verify with a truth table.
- Energy-saving through zone control.

LIMITATIONS

- Authorised codes are fixed in hardware; changing them needs rewiring.
- No memory, so wrong attempts cannot be counted and there is no lockout.
- Only 16 code combinations; switches can be observed or guessed.
- Basic model controls a single door and a simple zone.

APPLICATIONS & FUTURE SCOPE

- Homes, offices, laboratories and server rooms.
- Educational demonstration of combinational logic design.
- Future scope: keypad encoder, attempt lockout using flip-flops, longer codes.
- Can be upgraded with Arduino, RFID or IoT for smart access control.

CONCLUSION

- A digital door access system was designed and simulated using only combinational logic.
- The 4-bit code, Enter key and zone inputs were combined with AND, NOT and OR gates to control the lock, denied indication and zone power.
- The truth table and Logisim simulation agreed, so the design works as intended.
- Boolean simplification reduced the unlock logic to a single AND gate.

GITHUB REPOSITORY

- Repository Link: https://github.com/your-username/digital-door-access-system
- Contents: circuit diagram, Logisim files, report, PPT, poster, and source files.
- README includes project description, components, working, and setup steps.

THANK YOU

Questions & Discussion

